

1. Administrative & Study Identification

Full Study Title: A Study to Describe the Characteristics, Health Care Interventions and Outcomes of Patients With Major Bleedings in the Presence of Factor Xa Inhibitor Treatment (REVERXaL) **Short Title/Acronym:** REVERXaL **Protocol Number:** D9603R00003 **NCT Number:** NCT06147830 **Sponsor Name:** AstraZeneca **Investigational Product:** N/A (Observational study evaluating standard Factor Xa inhibitor reversal/replacement agents) **Phase of Development:** Phase 4 (Observational) **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To describe patient characteristics, health care interventions during the acute-care phase, in-hospital outcomes, and associations between timing of reversal/replacement agent administration and in-hospital outcomes in patients hospitalized with Factor Xa inhibitor (FXai)-associated major bleeds. **Secondary Objectives:** To describe clinical outcomes and patient-reported outcome measures (PROMs) including SF-36 v2 and EQ-5D-5L at 30 and 90 days, and evaluate the timing of administration of reversal/replacement therapy.

2.2 Background & Rationale Direct oral anticoagulants, including factor Xa inhibitors (FXai), are replacing vitamin K antagonists as the most commonly prescribed treatment for reducing the risk of thrombotic events. While the risk of FXai-associated spontaneous bleeds is established, less is understood about their management and the effect of treatment on clinical and patient-reported outcomes in a real-world setting. This study generates real-world evidence to inform clinical guidelines, health system decision-making, and streamline treatment pathways in this population.

3. Study Design & Methodology

3.1 Design Framework Study Type: Observational (Multinational, Longitudinal) **Control Type:** Historic cohort (Cohort A) and Prospective cohort (Cohort B) **Blinding:** Open-label / None **Randomization:** N/A

3.2 Planned Enrollment & Duration Target **\$N\$:** 2,202 patients **Number of Sites:** Multinational (Germany, Japan, United Kingdom, and the United States) **Estimated Study Start:** 04-Dec-2023 **Estimated Primary Completion:** 30-Nov-2024

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Aged ≥ 18 years on the index date. 2. Admitted to the hospital with an acute major bleeding or developed an acute bleeding while already in hospital. 3. Ongoing treatment with a Factor Xa inhibitor before the index date (Additionally for Cohort B: Administered reversal or replacement therapy).

4.2 Exclusion Criteria 1. Pregnant women. 2. Patients enrolled in any interventional trial that includes reversal/replacement agents. 3. Inability to provide signed and dated informed consent or obtain a waiver.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: N/A (Observational). Patients receive routine medical care and reversal/replacement therapies (e.g., prothrombin complex concentrate) as determined by their treating physicians. **Route of Administration:** As per routine clinical care (e.g., IV). **Duration of Treatment:** Acute hospital care phase with outcomes tracked up to hospital discharge (Cohort A) and up to 90 days post-discharge (Cohort B).

5.2 Concomitant & Prohibited Therapy Permitted: Standard medical care, standard clinical assessments, and procedures as prescribed by the treating physician for major bleeding management. **Prohibited:** Concurrent participation in an interventional trial testing experimental reversal/replacement agents.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	Screening / Index	Day 0 Hospital Admission/ Bleed Onset: Informed Consent, Medical History, Bleeding Presentation, Index Date Definition
2	Interim (Acute)	In-Hospital Routine Clinical Care: Administration of reversal/ replacement therapies, acute-phase health interventions, Safety/ AE assessments
3	End of Study	Days 30 & 90 Discharge outcomes, Clinical Assessments, PROMs (SF-36 v2, EQ-5D-5L) for Cohort B

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Descriptive patient characteristics, clinical presentation, and health care interventions provided during the acute care phase. **Measurement**

Method: Secondary data collection via extraction from electronic/paper medical records into electronic case report forms (eCRFs).

7.2 Secondary & Exploratory Endpoints 1. In-hospital clinical outcomes and their associations with the timing of reversal/replacement agent administration. 2. Patient-reported outcome measures (PROMs) including physical and mental health dimensions evaluated via the 36-Item Short Form Health Survey Version 2 (SF-36 v2) and EQ-5D-5L at 30 and 90 days.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via routine clinical visits. Investigators evaluate medical history, bleeding presentation, and any AE/SAE experiences since the previous clinical visit. **Serious Adverse Events (SAEs):** Captured through routine care observations and extracted to the eCRF. **Data Monitoring Committee (DMC):** N/A (Observational, non-interventional study).

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Historic Cohort A (up to hospital discharge) and Prospective Cohort B (up to 90 days post-treatment). **Sample Size Justification:** Approximately 2,202 patients; descriptive in nature to outline real-world population profiles. **Handling of Missing Data:** The study is descriptive without pre-specified statistical hypotheses. The baseline value for analyses is defined as the last non-missing observed measurement prior to admission or bleed onset.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Pharmacoepidemiology Practices, Good Clinical Practice (GCP), and the Declaration of Helsinki. **Monitoring Plan:** Data abstraction from primary (patient-reported) and secondary (medical records) sources during routine visits. **Audit Trail:** Electronic case report forms (eCRFs) utilized for consistent tracking of clinical pathway details and protocol adherence.

11. Study Identifiers & Governance

Primary ID: {{D9603R00003}} **Registry Number:** {{NCT06147830}}
Sponsor/Lead Organization: {{AstraZeneca}} **Study Title:** {{REVERXaL}} **Official Regulatory Title:** {{A Study to Describe the Characteristics, Health Care Interventions and Outcomes of

12. Clinical Framework (Study Specific)

Primary Condition: {{Major bleeding associated with Factor Xa inhibitors}} **Diagnosis Threshold:** {{Acute major bleeding requiring hospitalization or occurring in-hospital}} **Medical Methodology:** {{Routine standard medical care and reversal/replacement therapies}} **Optimization Target:** {{Real-world bleeding control and health outcome characterization}}

13. Study Procedures & Methodology

Management Pathway: {{Routine medical care pathway for acute major bleeding}} **Education Protocol (HCP):** {{N/A - Observational study design}} **Education Protocol (Patient):** {{N/A - Observational study design}} **Quality Audit Mechanism:** {{Secondary data extraction and eCRF validation}}

14. Observation & Follow-Up Schedule

Total Duration: {{Up to 90 days post-index date for Cohort B}} **Onsite Visit Cadence:** {{Index hospitalization and routine clinical follow-ups}} **Remote Monitoring Frequency:** {{Assessments aligned with routine care at 30 and 90 days}} **Checklist Review Interval:** {{N/A}}

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				